GOVERNMENT OF KHYBER PAKHTUNKHWA

MedLedger

Hospital Pharmacy Management System & Sehat Card Sahulat Program

District Headquarter (DHQ) Hospital Kohat

🚀 100% FEFO Adherence

Automated First-Expiry-First-Out dispensing algorithms ensure zero stock expiration losses in hospital wards.

🪪 Sehat Card Integration

Real-time CNIC claims validation, transparent package coverage checks, and instant government reimbursement filing.

⚡ Offline Resilience & PWA

PWA offline caching for OPD counters during network interruptions, with seamless background auto-sync.

📘 About This User Manual

This official visual operational guide provides pharmacy managers, staff pharmacists, claims verification officers, and clinical administrators with comprehensive instructions on utilizing all features of the MedLedger system. Each section features live system screenshots, annotated control points, and step-by-step procedures.

FACILITY	SYSTEM VERSION	LIVE DEPLOYMENT	RELEASE DATE
DHQ Hospital Kohat	v0.1.0 Enterprise	medledger-tau.vercel.app	October 2026

📑 Table of Contents

SYSTEM MAP

1. Global Navigation & Live Header — Page 3
2. Real-Time Update Notifications & PWA — Page 4
3. Pharmacy Operations Dashboard — Page 5
4. Patient Registry & CNIC Lookup — Page 6
5. Inpatient Admissions & Ward Beds — Page 7
6. Electronic Prescriptions Intake — Page 8
7. Dispensary & FEFO Fulfillment — Page 9
8. Medicines Catalogue & Pricing — Page 10
9. Inventory Control & Batch Expiry — Page 11
10. Supplier Management & Vendor Directory — Page 12
11. Purchase Orders & Goods Receiving (GRN) — Page 13
12. KP Sehat Card Claims & Billing — Page 14
13. Financial Reports & Stock Reconciliation — Page 15
14. Security, Roles & Daily Operational Routine — Page 16

🔑 User Roles in MedLedger

MedLedger implements Role-Based Access Control (RBAC) configured for DHQ Hospital Kohat hierarchy:

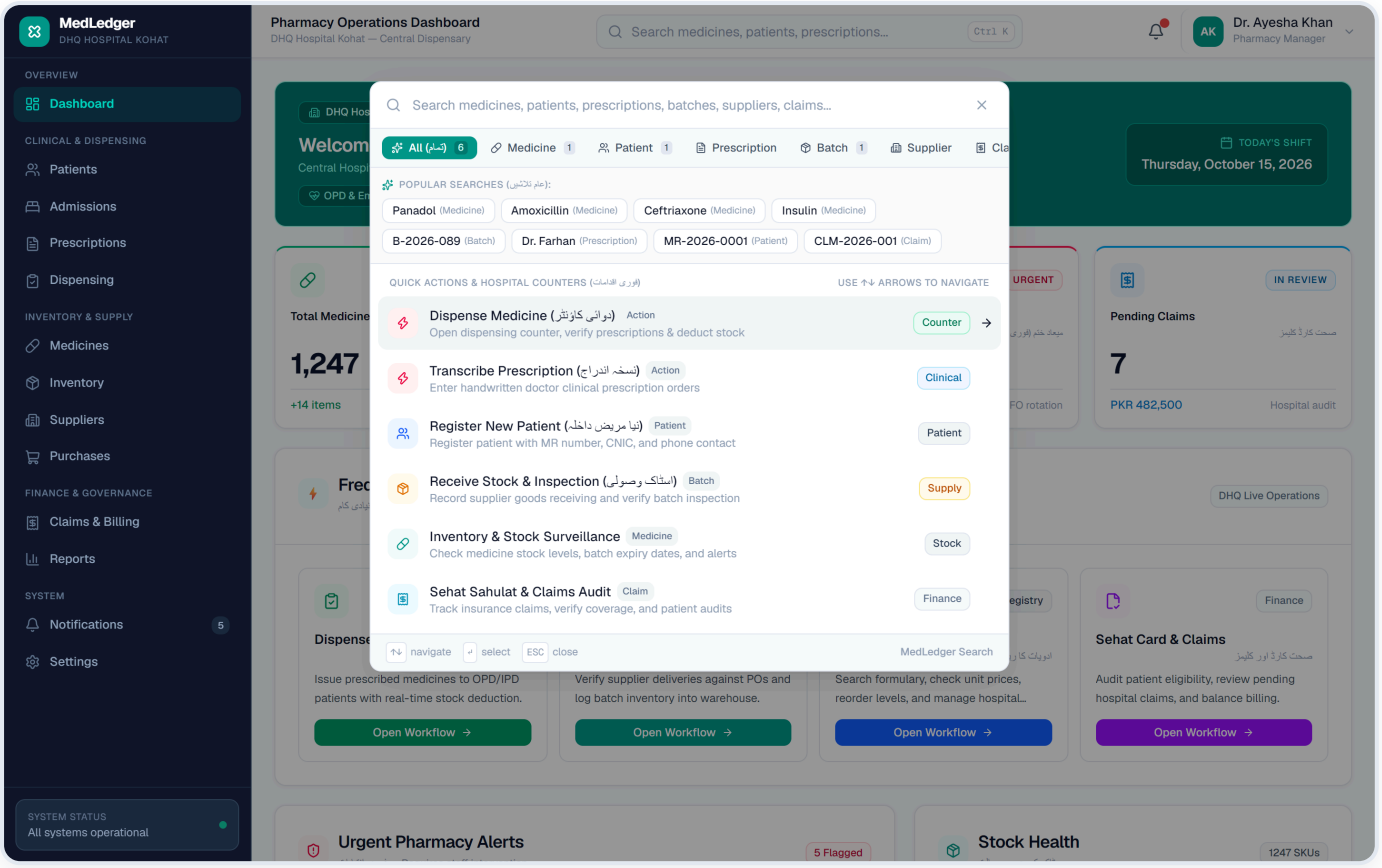
- **Staff Pharmacist (Ayesha Khan)**: Frontline OPD/IPD dispensing, prescription verification, patient lookup, and stock adjustments.
- **Pharmacy Manager**: Purchase order approval, supplier relations, batch inventory receiving, and audit oversight.
- **Claims Verification Officer**: KP Sehat Card eligibility verification, claim dispute handling, and reimbursement submission.
- **Chief Medical Officer (CMO)**: Hospital-wide executive analytics, budget compliance, and stock reconciliation audits.

1. Global Header & Fast Navigation (Ctrl + K)

CORE INTERFACE

Universal Search & Operations Bar

The top header stays pinned across all screens, granting instant access to hospital shift data, global medicine search, operational alerts, and role simulation.



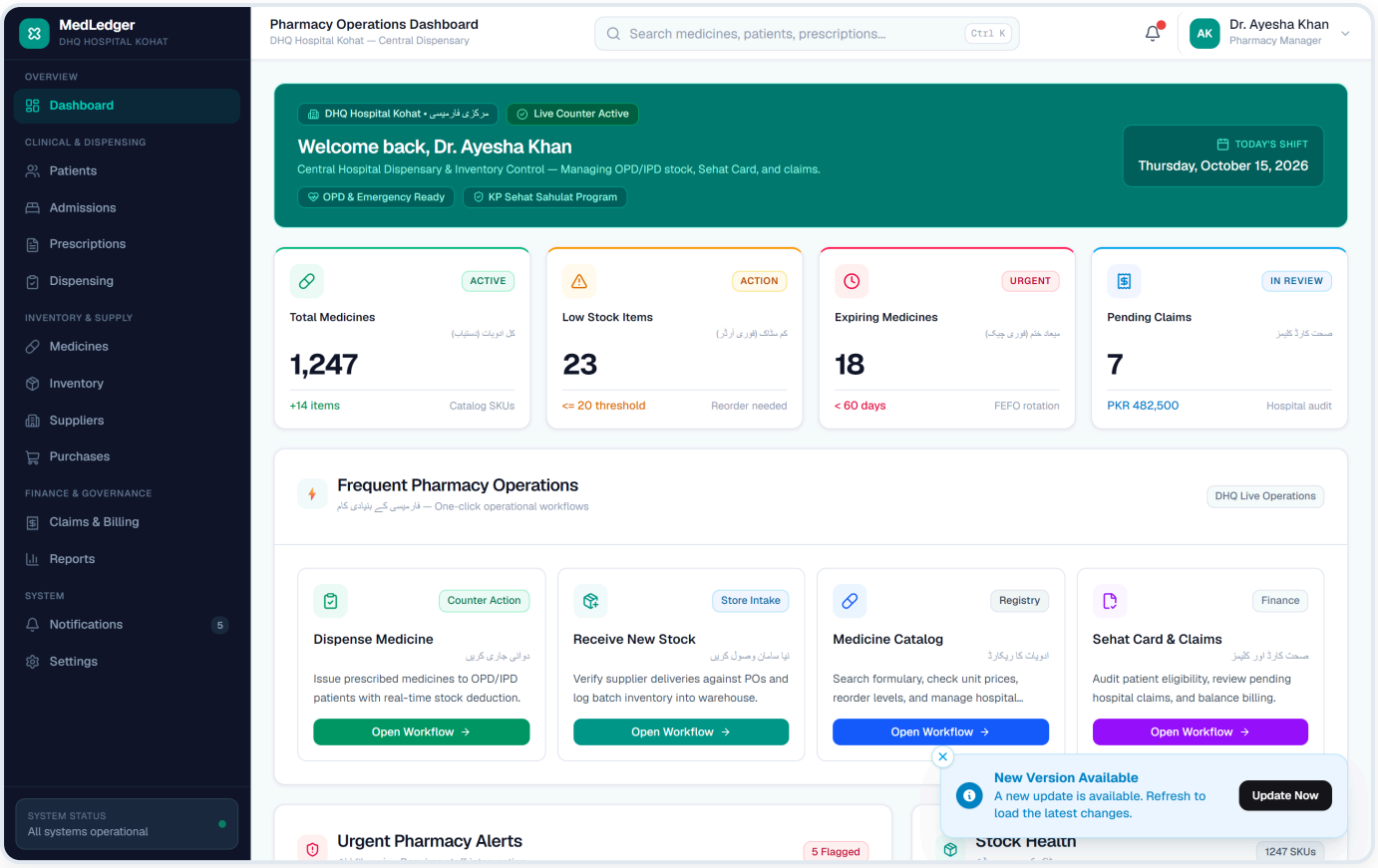
Feature Element	Visual Location	How to Use
Search Shortcut (Ctrl + K)	Center of Header / Shortcut key	Press `Ctrl + K` from anywhere in the app to open the instant search modal. Type medicine brand, generic name, or patient CNIC.
Quick Action Shortcuts	Inside Search Palette Dialog	Click predefined shortcuts like *"Dispense Medicine"*, *"Add Patient"*, or *"Review Claims"* for single-click navigation.
Active Hospital Shift	Left Header subtitle	Displays current dispensary station: *DHQ Hospital Kohat — Central Dispensary*.
Simulated Role Switcher	Top Right Avatar (AK)	Click the user profile dropdown to preview system permissions as Staff Pharmacist, Pharmacy Manager, or Claims Officer.

2. Real-Time Update Notifications & PWA

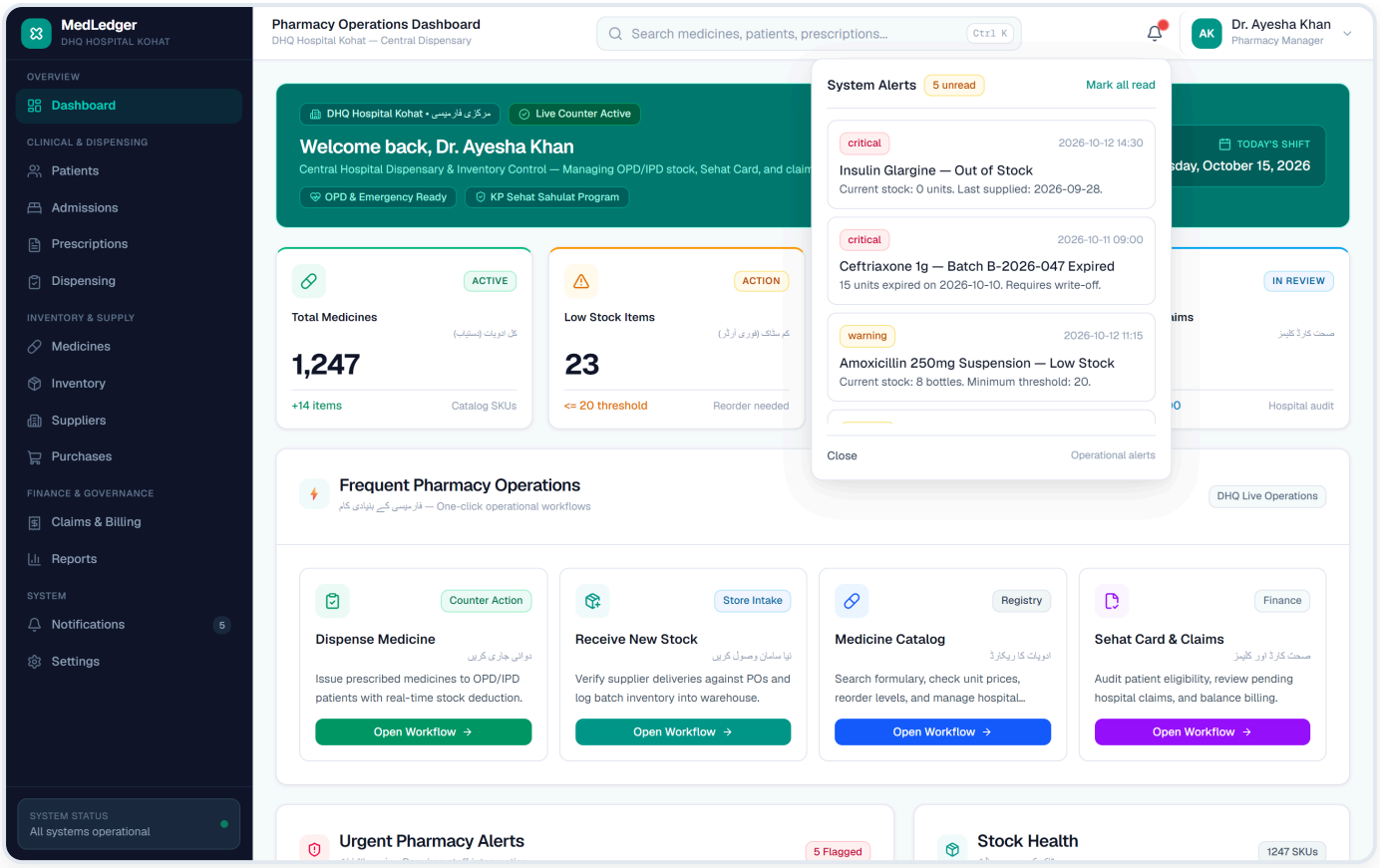
SYSTEM RELIABILITY

Floating "New Version Available" Alert & Bell Notifications

MedLedger actively monitors deployment versions and hospital operational alerts so pharmacists never dispense with stale rules.



Feature Element	Visual Characteristics	Operational Function
New Version Available Toast	Bottom-Right floating modal (SKY BLUE)	Appears automatically within 60 seconds whenever an IT deployment or pricing update occurs on Vercel.
"Update Now" Button	High-contrast black button on right	One-click activation: signals the waiting Service Worker to take over and reloads the page with zero data loss.
Dismiss Circle (x)	Top-Left white circular icon	Allows postponing the refresh if the pharmacist is currently in the middle of typing an active prescription line.



🔔 How to Handle System Alerts:

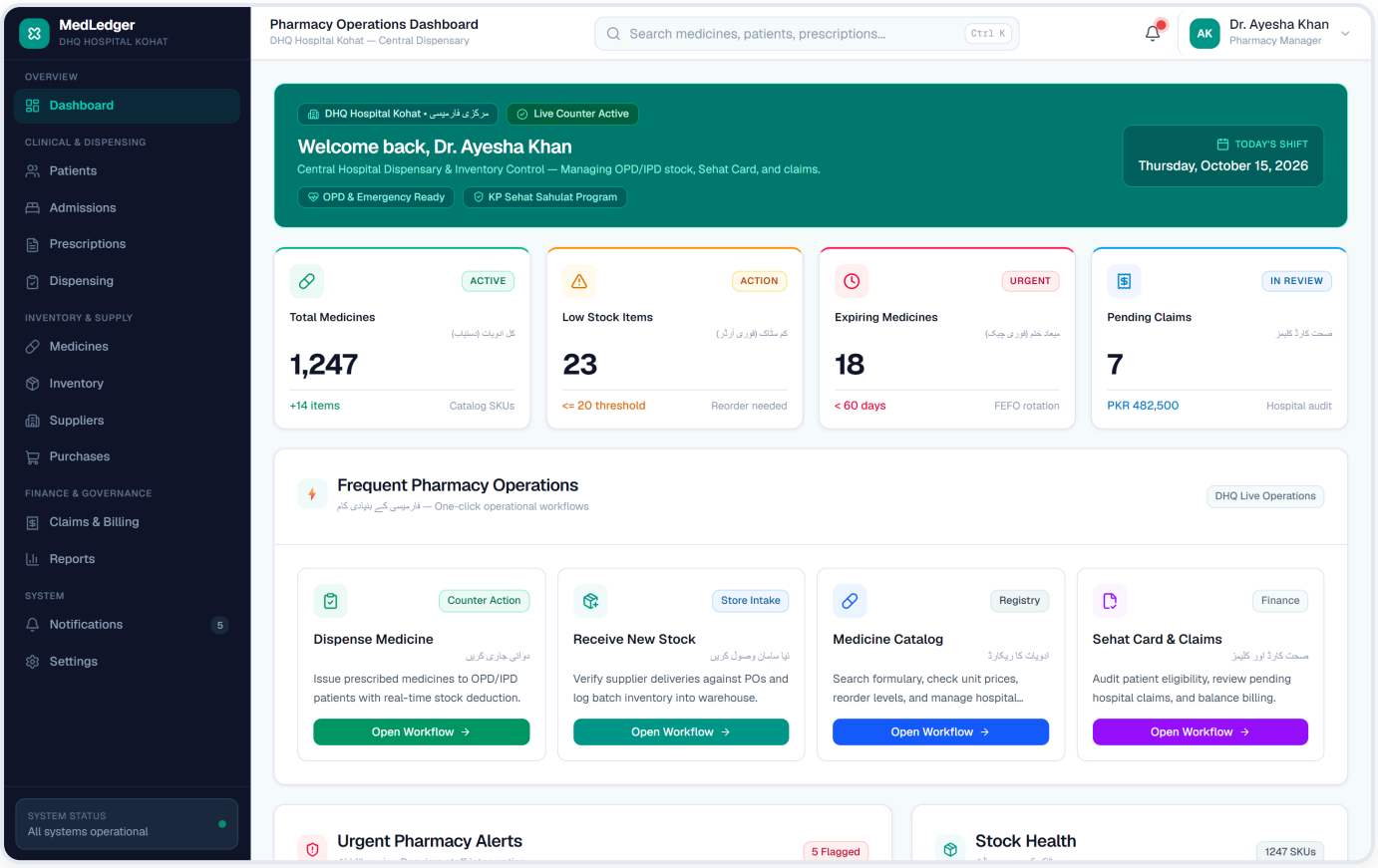
1. Click the **Bell Icon** in the top header (with pulsating red dot when unread alerts exist).
2. Review high-priority alerts: Near-expiry antibiotics, low insulin stock, and rejected Sehat Card claims.
3. Click **"Mark all read"** once actions have been delegated to shift pharmacists.

3. Pharmacy Operations Dashboard

SHIFT CONTROL CENTER

Live Operations Overview

The central mission control for DHQ Hospital Kohat, providing real-time stock surveillance and Sehat Card claim KPIs.



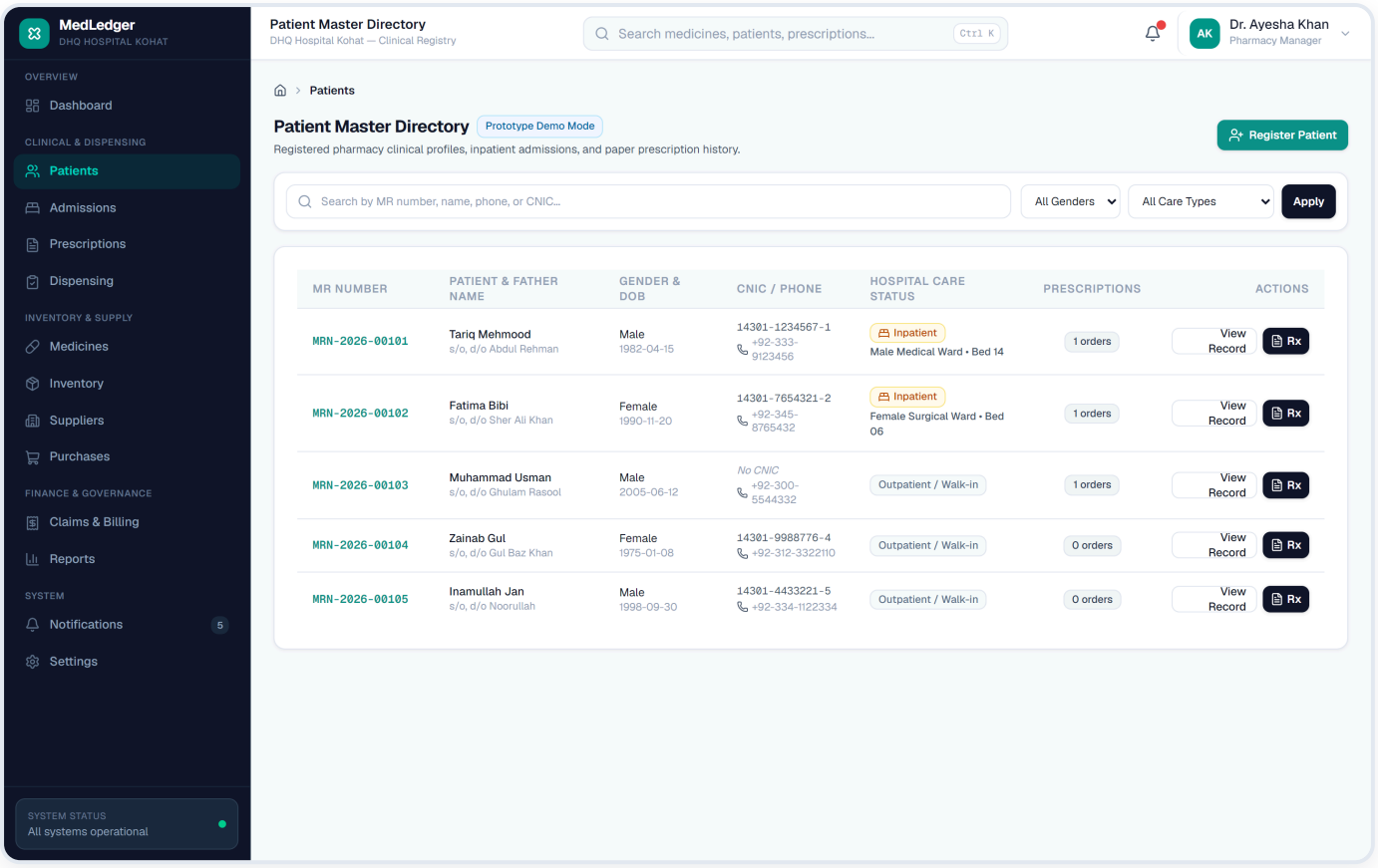
Metric Card	Current State	Significance & Action
Total Medicines	1,247 SKUS	Total active catalog formulations approved for DHQ Hospital formulary.
Low Stock Items	23 ITEMS	Items below the minimum hospital safety buffer threshold (≤ 20 units). Requires raising a Purchase Order.
Expiring Medicines	18 BATCHES	Batches expiring within the next 90 days. Prioritized by automated FEFO algorithm to minimize expiration loss.
Pending Claims	7 IN REVIEW	Sehat Card Sahulat claims currently being vetted by the state insurance panel for hospital reimbursement.

4. Patient Registry & Sehat Card Lookup

CLINICAL RECORDS

Patient Database & National ID Verification

Centralized directory linking patients with CNIC, Sehat Card entitlement, and OPD/IPD admission histories.



📋 Patient Intake & Sehat Card Verification Workflow:

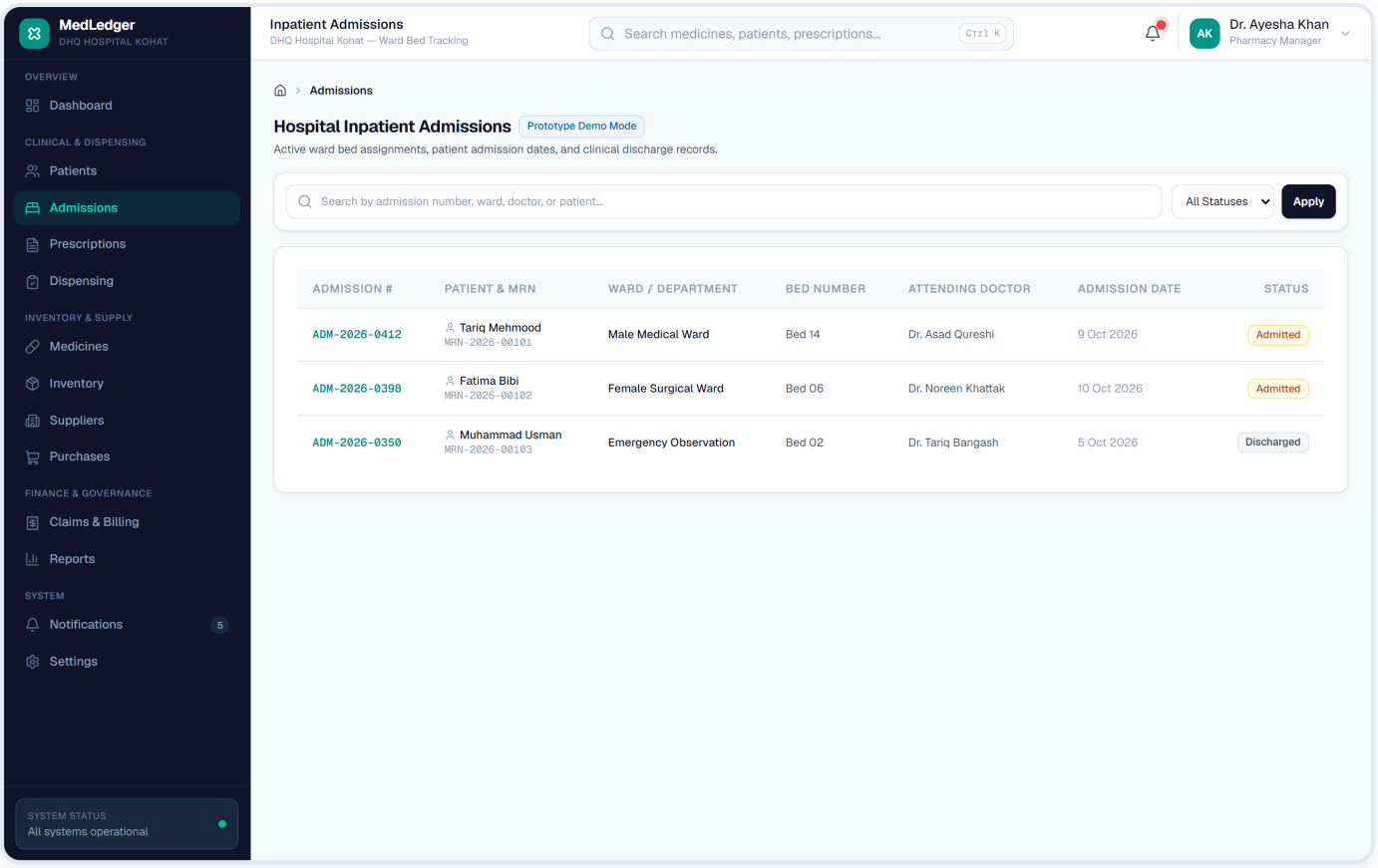
1. Navigate to **Patients** via the left sidebar.
2. Enter the patient's 13-digit National ID (CNIC) or Medical Record Number (MRN) into the search filter.
3. Observe the **Sehat Card Status badge**:
   - VERIFIED : Patient has active government medical coverage. Free medicine eligible.
   - PENDING VERIFICATION : Requires identity confirmation with NADRA / Sehat Card desk.
4. Click **"+ New Patient"** in the top right to register emergency or walk-in patients.

5. Inpatient Admissions & Ward Management

IPD OPERATIONS

Ward Bed Allocation & Daily Bedside Dispensing

Track admitted patients in Surgical, Medical, Pediatric, and ICU wards at DHQ Hospital Kohat.



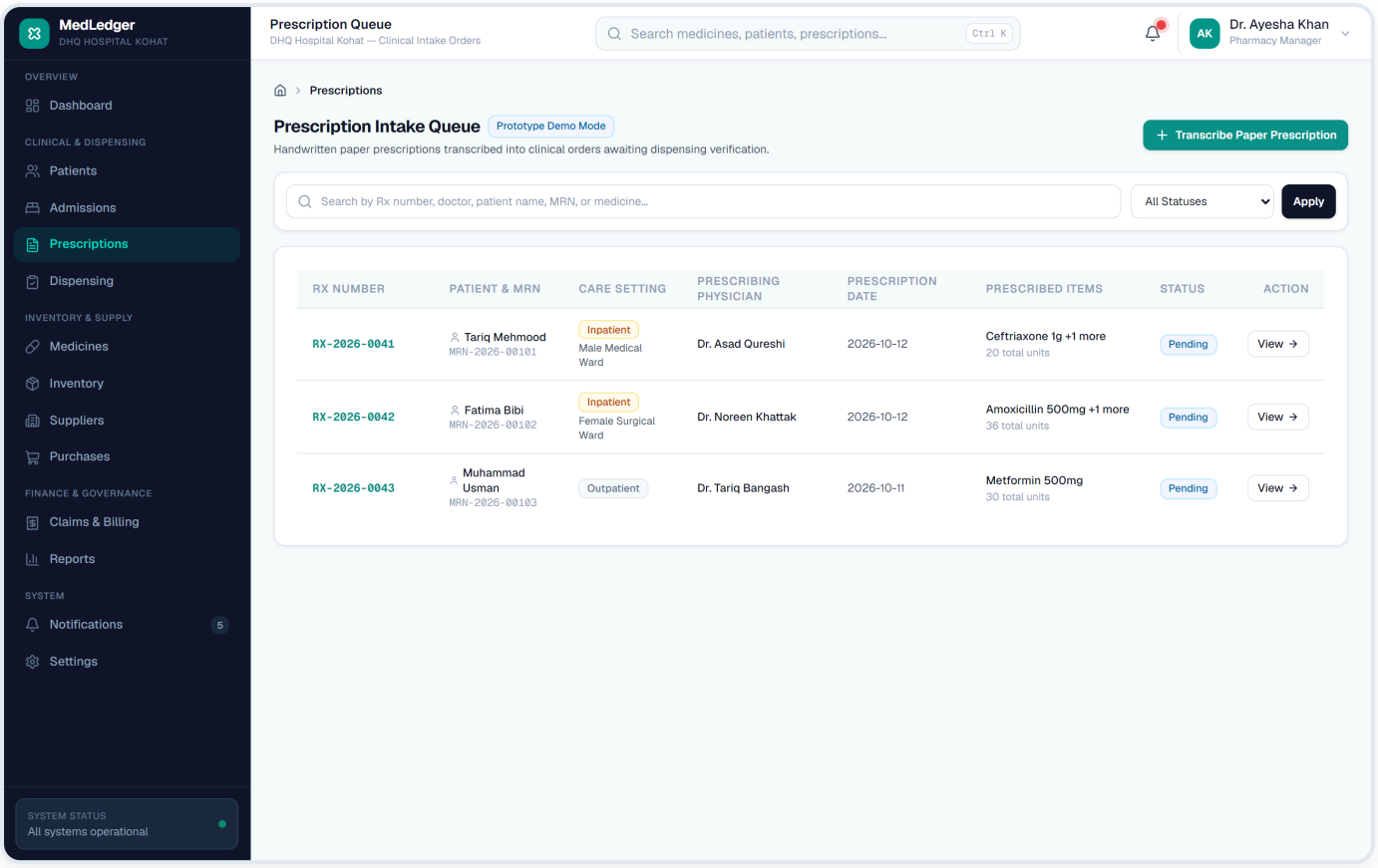
Column	Description	Operational Instructions
Admission ID & Ward	Ward designation (e.g. Ward 3B, Bed 12)	Determines where dispensary delivery runners deliver scheduled intravenous and oral medications.
Admitting Doctor	Attending Physician / Specialist	Only authorized prescriptions signed by the registered attending consultant can be charged to Sehat Card.
Status	ADMITTED / DISCHARGED	Upon patient discharge, all pending ward medicine balances are reconciled before final bill clearance.

6. Electronic Prescriptions Intake

PHYSICIAN ORDERS

Digital Doctor Orders & Medication Schedules

Eliminates handwritten medication errors with structured dosage, frequency, and duration tracking.



💊 Prescriptions Verification Workflow:

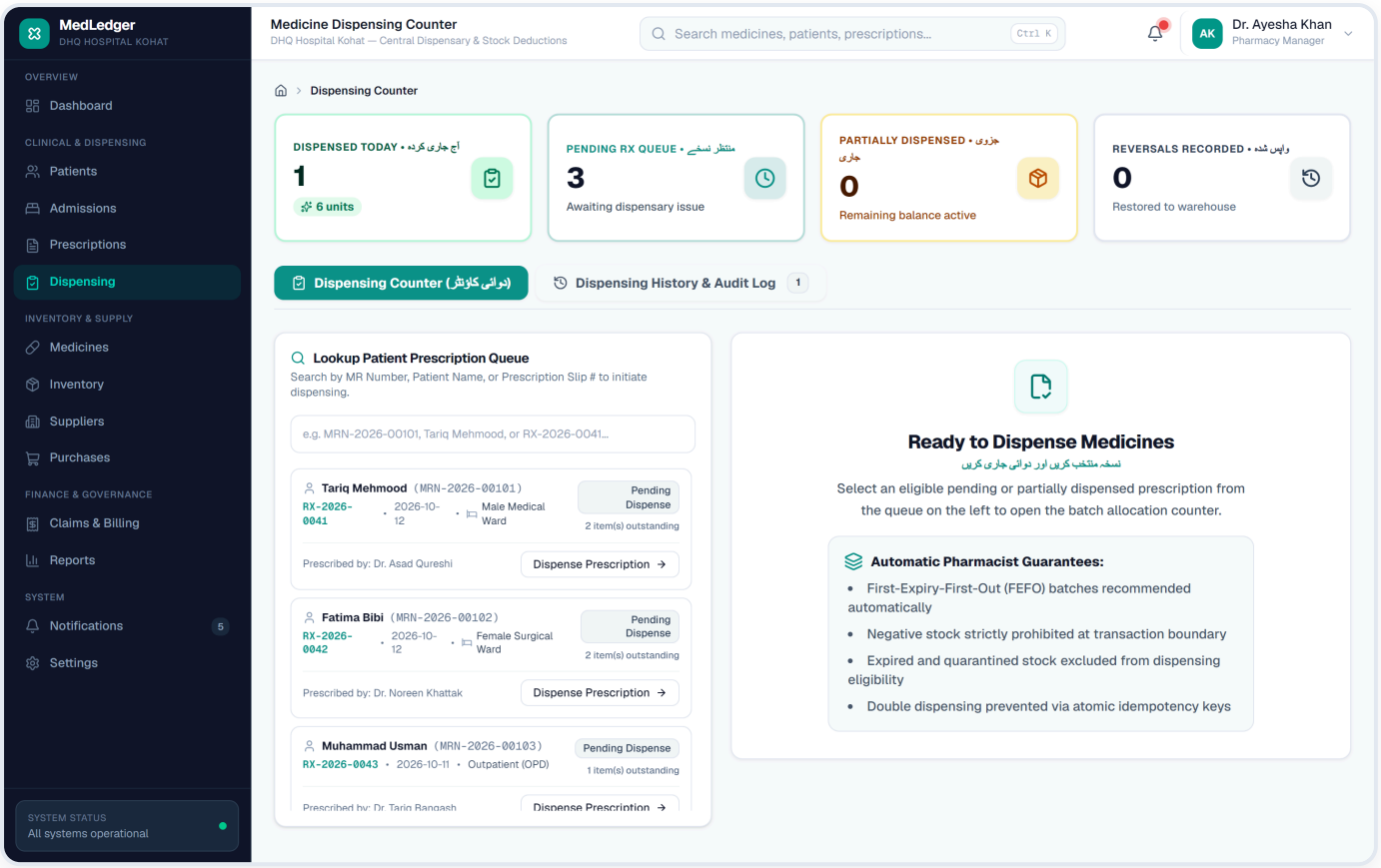
1. Click on any prescription row to inspect individual drug lines, dosage (e.g. *500mg BID*), and route.
2. Review ICD-10 diagnosis tagging to ensure treatment matches government hospital guidelines.
3. Check the **Fulfillment Status**:
   - PENDING : Ready for dispensary pharmacist verification.
   - DISPENSED : Medications handed over to patient/attendant.
4. Click **"Dispense Medicine"** to transfer the prescription directly into the FEFO dispensing engine.

7. Dispensary & FEFO Fulfillment (First-Expiry-First-Out)

PRIMARY WORKFLOW

Automated Expiry-Sorted Dispensing Engine

Guarantees optimal drug stock rotation by automatically selecting earliest-expiring active batches.



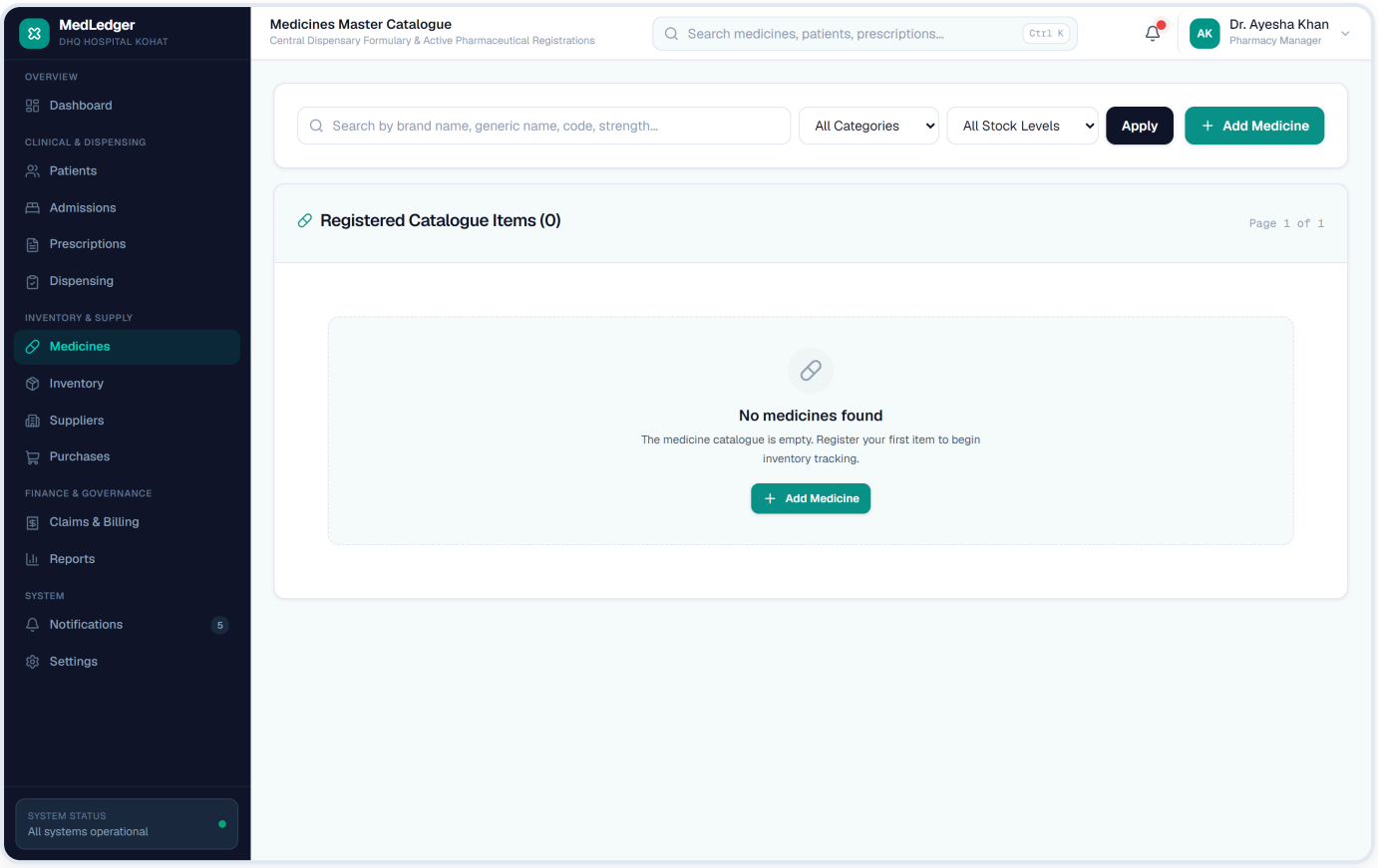
Step	Action	System Protection Rule
1. Select Prescription	Load patient prescription by token or MRN	System locks medicine inventory to prevent double-dispensing across different dispensary counters.
2. Automated Batch Selection	System selects batch with earliest expiration date	**FEFO Enforced:** Pharmacists cannot bypass earlier batches without manager override reason.
3. Claim / Cash Allocation	Calculate Sehat Card coverage vs patient copay	Exact integer paisa precision arithmetic ensures ledger accuracy with zero rounding drift.
4. Print Bilingual Receipt	Generate thermal receipt in Urdu & English	Includes dosage instructions in Urdu for clear patient comprehension.

8. Medicines Catalogue & Pricing Control

DRUG MASTER

Hospital Formulary & Dosage Forms

Master pharmaceutical catalogue compliant with DRAP (Drug Regulatory Authority of Pakistan) standards.



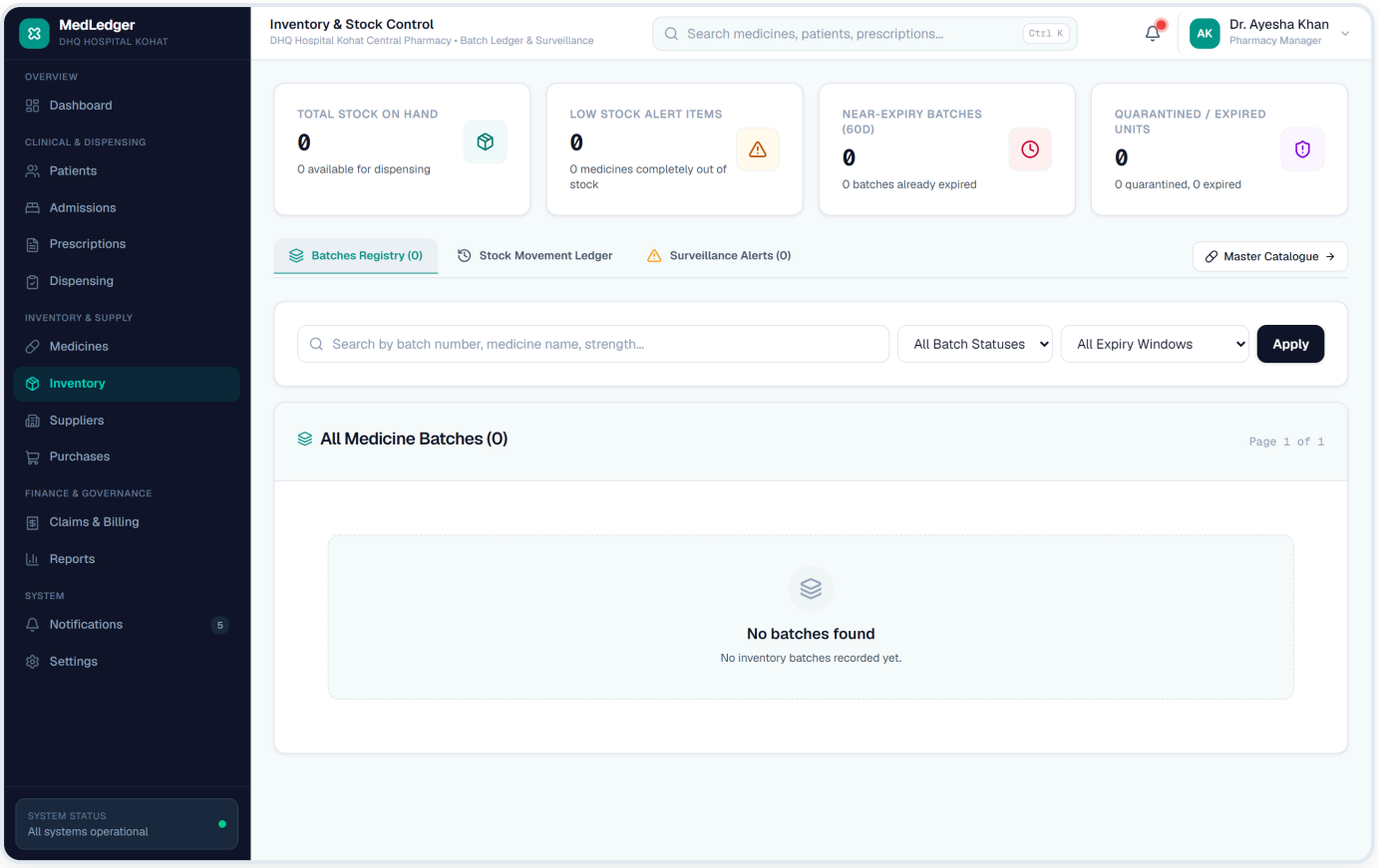
💡 **Catalogue Management Standard:** Every medicine entry contains both its commercial brand name (e.g., *Augmentin*) and International Nonproprietary Name generic title (*Amoxicillin + Clavulanic Acid*), strength (*625mg*), dosage form (*Tablet*), and maximum retail price (MRP).

9. Inventory Control & Batch Expiry Surveillance

STOCK MANAGEMENT

Batch-Level Tracking & Safety Stock Triggers

Real-time inventory levels broken down by batch code, supplier provenance, and exact storage location.



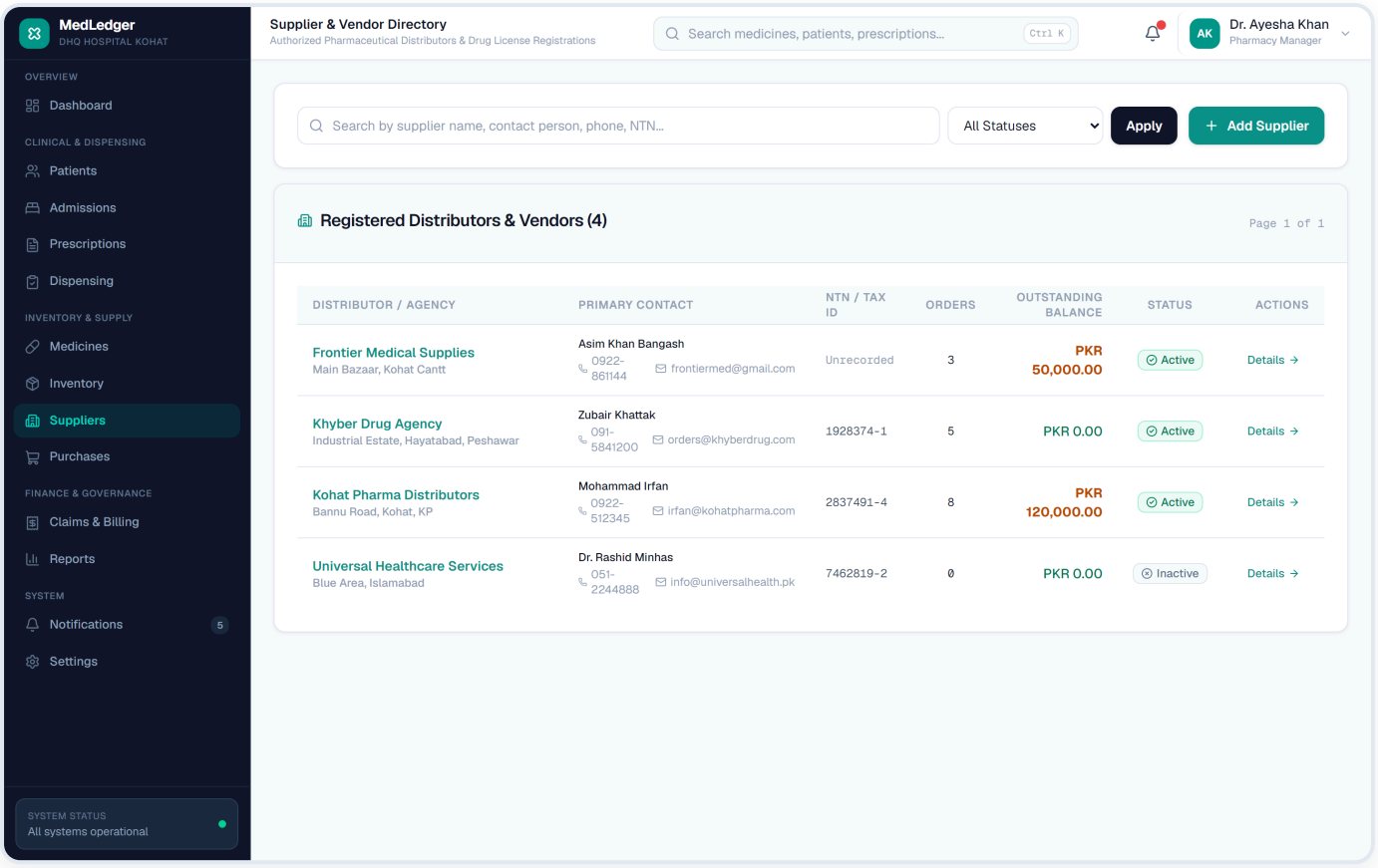
Indicator	Color Status	Immediate Mandatory Action
Adequate Stock	IN STOCK	Inventory level exceeds 30-day average consumption. Normal dispensing.
Low Stock Warning	REORDER LEVEL	Quantity is at or below reorder threshold. Triggers automated procurement requisition.
Critical Expiry Alert	EXPIRING < 90 DAYS	Move stock to high-volume outpatient OPD counter to guarantee usage before expiry.

10. Supplier Management & Vendor Directory

PROCUREMENT

Approved Pharmaceutical Distributors

Pre-qualified drug distributors supplying DHQ Hospital Kohat with active drug sales licenses.



🏢 Vendor Compliance Requirements:

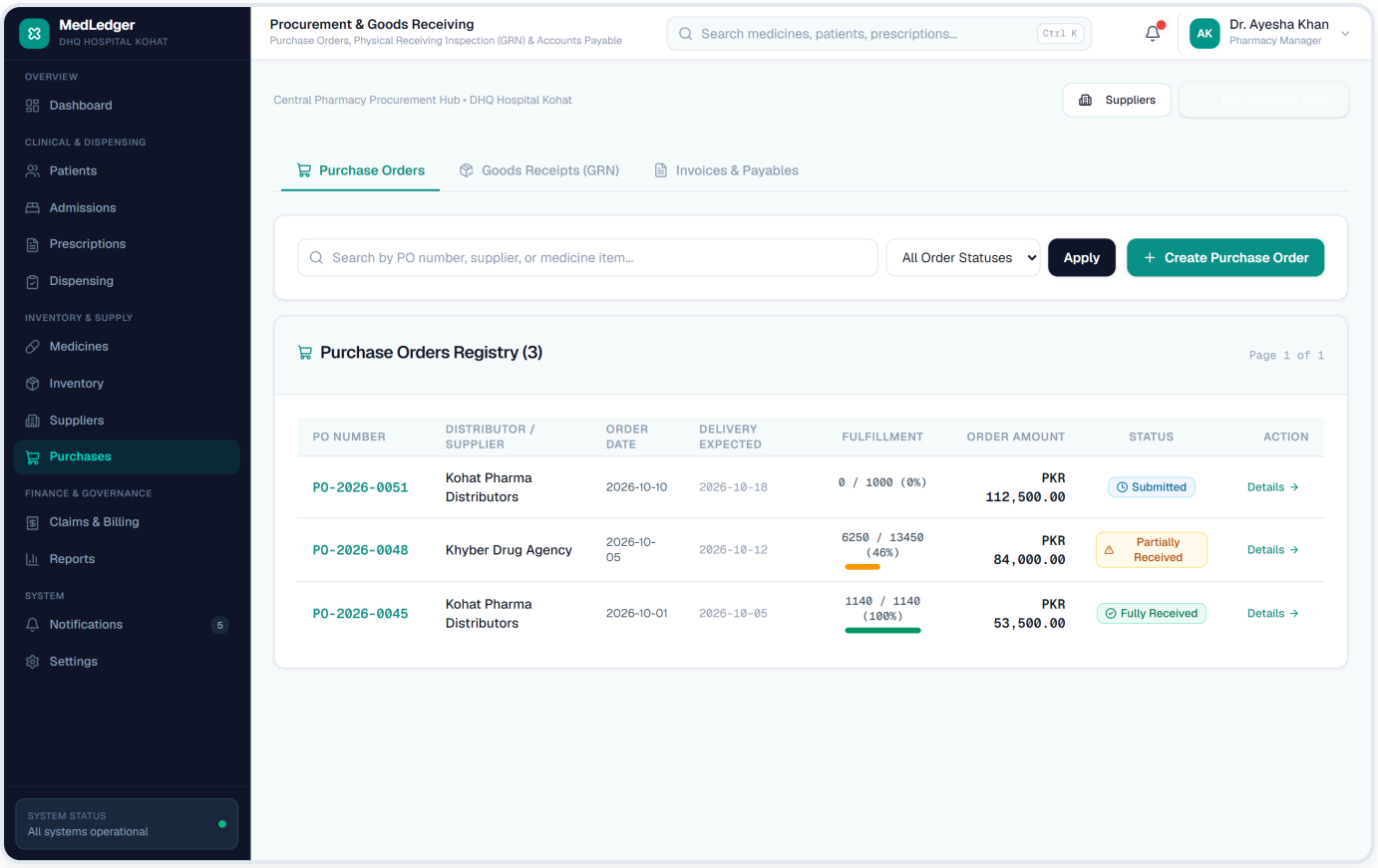
- **Drug Sale License (DSL)**: Must be validated and stored on file with expiry dates.
- **NTN & Sales Tax Registration**: Verified with FBR for government withholding tax deduction.
- **Cold Chain Certificate**: Mandatory for insulin, vaccines, and biologics distributors.

11. Purchase Orders & Goods Receiving Notes (GRN)

SUPPLY CHAIN

Inward Batch Logging & Quality Verification

Three-way matching between Purchase Order (PO), Supplier Invoice, and physical Goods Receiving Note (GRN).



📦 Step-by-Step Receiving Protocol:

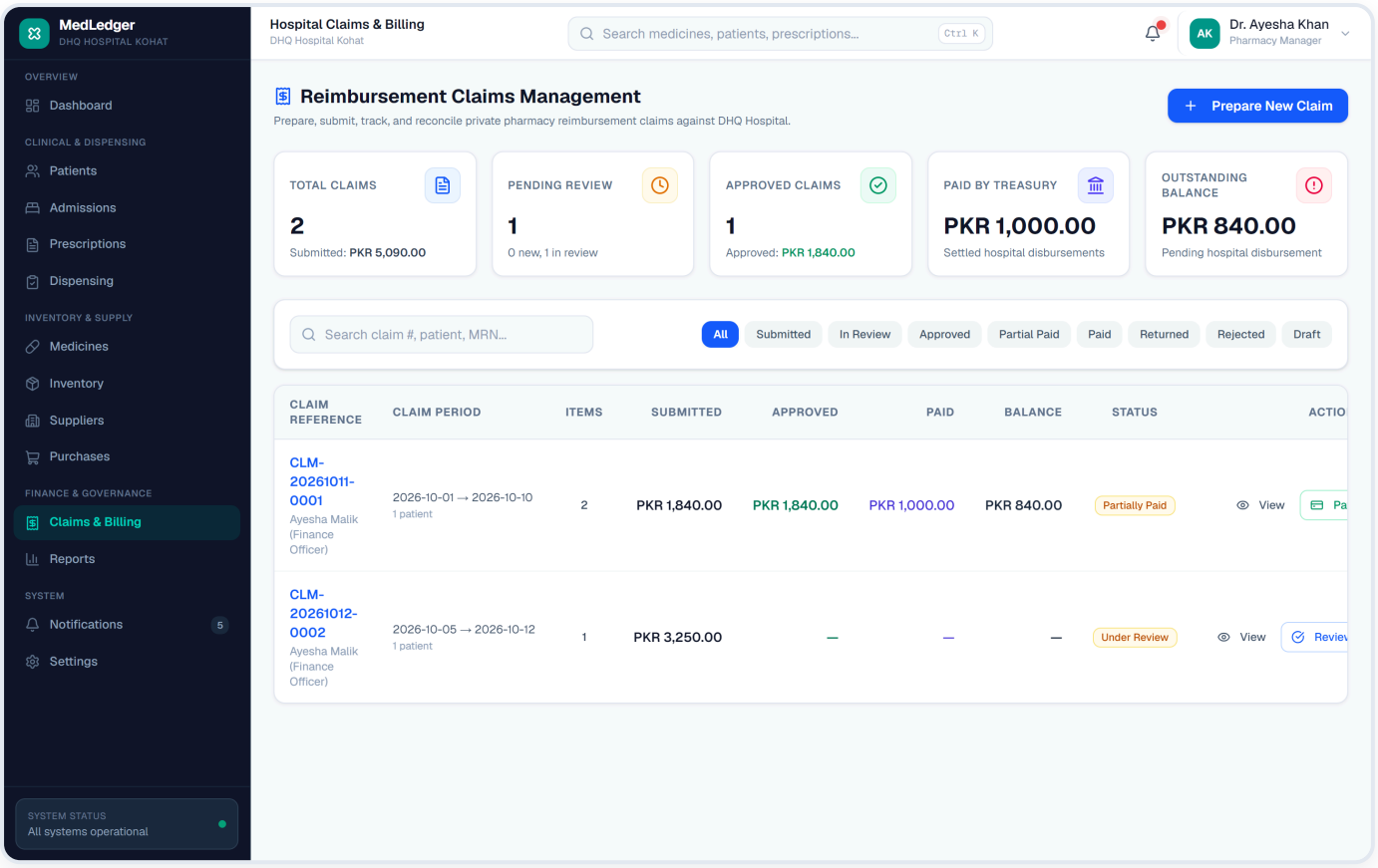
1. Inspect physical medicine cartons for tamper seals, batch numbers, and manufacturer expiry stamps.
2. Open MedLedger → **Purchases** → Locate the matching Purchase Order.
3. Click **"Log Goods Receipt (GRN)"**. Enter actual received pack quantities and batch lot numbers.
4. Verify unit purchase price against invoice. MedLedger automatically increases stock upon submission.

12. KP Sehat Card Sahulat Claims & Billing

GOVERNMENT PROGRAM

Universal Health Coverage Reimbursements

Automated claims generation from verified dispensing transactions for government insurance reimbursement.



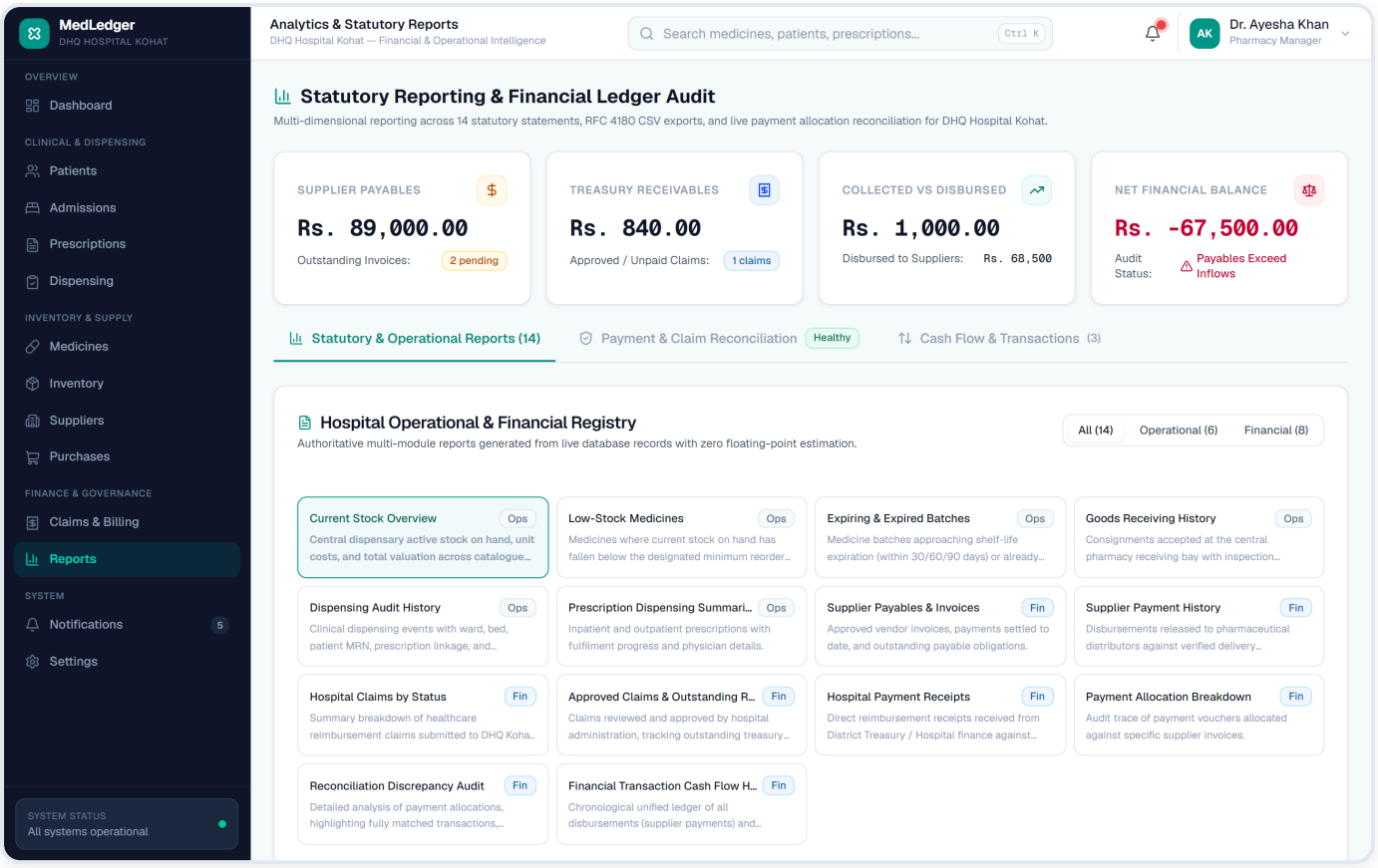
Claim Status	Color Badge	Meaning & Next Step
Draft / Pending	UNDER REVIEW	Claim generated by pharmacy dispensary. Awaiting internal claims officer audit check.
Submitted	SUBMITTED	Dispatched electronically to KP Sehat Card TPA portal for formal review.
Approved & Settled	SETTLED / PAID	Funds transferred directly into DHQ Hospital Kohat designated bank account.
Rejected / Disputed	REJECTED	Requires attaching additional clinical notes or corrected admission ICD-10 code.

13. Financial Reports & Stock Reconciliation

GOVERNANCE & AUDIT

Executive Analytics & Exportable Audit Trails

Comprehensive reports for hospital financial audits, inventory valuation, and consumption forecasting.



📊 Key Reporting Capabilities:

- **Daily Dispensary Cash vs Sehat Card Summary**: Reconciles cash drawer against insurance tallies.
- **Stock Valuation by Batch**: Calculates total hospital capital locked in medicines at landed cost.
- **Expiry Loss Prevention Audit**: Identifies batches requiring transfer before expiration date.
- **CSV / Excel Export**: Single-click export for provincial health ministry reports.

14. Security, Roles & Daily Operational Routine

HOSPITAL SOP

Standard Operating Procedure (SOP) for Daily Shifts

Mandatory daily routine for morning, evening, and night dispensary shifts at DHQ Hospital Kohat.

Shift Phase	Primary Tasks	System Verification
Morning Handover (08:00)	Log in with personal staff credentials. Inspect Priority Alerts on Dashboard. Review Low Stock warnings.	Verify physical emergency drawer stock matches MedLedger reported count.
Mid-Shift Dispensing	Fulfill OPD and IPD ward prescriptions strictly using FEFO suggestions. Check Sehat Card validity.	Always verify patient CNIC matches card holder before dispensing government-subsidized items.
Evening Reconciliation (16:00)	Print shift cash summary from Reports. Handover unfulfilled ward orders to incoming night pharmacist.	Review pending claims to ensure zero backlogged Sehat Card tickets.

🛡️ Data Security & Integrity Policies:

- **Strict Audit Logging**: Every dispensing event, stock manual adjustment, and prescription modification records the pharmacist ID and timestamp.
- **Integer Paisa Arithmetic**: Financial calculations operate in paisa units (PKR * 100) to eliminate floating-point rounding discrepancies during accounting audits.
- **Row-Level Security (RLS)**: Clinical records are encrypted and protected by Supabase RLS database policies.

District Headquarter (DHQ) Hospital Kohat

MedLedger Pharmacy Management & KP Sehat Card Sahulat System

Technical Support: Tech4Edges | IT Directorate Health KP